Hardware CWE™ Special Interest Group (SIG)

Chair: Bob Heinemann (MITRE)

Co-Chair: “Manna” Parbati Kumar Manna (Intel)

MITRE Team: Steve Christey Coley, Alec Summers

MITRE

March 20, 2026

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Agenda

REMINDER: This meeting is being recorded.

1	RTL WG Update	Bob H	5 min
2	HW CWE Completeness Report Demo	Steve C	10 min
3	Enabling Comprehensive CWE-based Assurance for RISC-V Processors	Mark Labbato	30 min

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Housekeeping

- **Schedule:**
  - **Next Meeting April 17:**
  - **Meeting is now the third Friday each month**
    - **12:30 – 1:30 PM EST (16:30 – 17:30 UTC)**
    - **Microsoft Teams**
- **Contact: cwe@mitre.org**
- **Mailing List:** *hw-cwe-special-interest-group-sig-list@mitre.org*
- **Minutes from previous meetings available on our GitHub site:**
  - **https://github.com/CWE-CAPEC/hw-cwe-sig**

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Announcements

- **4.20 release planned for some time in April**
- **CWE Content Development Repository (CDR) is now fully public.**
  - Enables the broader community to view, track, and contribute to entries submissions.
  - Content suggestions begin with the CWE Submission Form:
    - https://cwe.mitre.org/community/submissions/overview.html
  - CDR can be accessed here:
    - https://github.com/CWE-CAPEC/CWE-Content-Development-Repository/

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RTL WG Updates

- **Discussion has broadened to include:**
  - RTL Bad coding practices
  - Simulation / synthesis mismatches
    - Buggy EDA Tool Optimizations
    - Certain RTL Behaviors
    - Design Decisions / Semantic Differences
  - RTL / HW Mismatches

- **CWE team still thinking through how we model in CWE**

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Prioritizing Completeness Improvements Readback

- **Q1: Should we choose a specific element, or a set of elements to prioritize for completeness?**
  - Demonstrative Examples, Observed Examples, Likelihood of exploit, & Detection Methods.
- **Q2: Which CWE(s) do we focus on?**
  - MIHW, Commonly Confused CWEs, “Coalition of the Willing”.
- **Q3: Should we focus more on depth than breadth?**
  - Depth; deeper examples and clearer descriptions would reduce ambiguity and improve consistent interpretation.
  - SystemVerilog and VHDL examples.
- **Q4: How can the HW-SIG help?**
  - Improving data quality and mapping consistency.
  - Engaging with hardware vendors (including through how CWEs are used in CVE contexts) and potentially offering SIG help when vendors are unsure.
  - Communicate CWE information more effectively to industry for awareness and alertness.

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HW CWE Completeness Demo

- **Steve Christey Coley**

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Enabling Comprehensive CWE-based Assurance for RISC-V Processors

- **Mark Labbato**

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Next Meeting (April 17)

CWE@MITRE.ORG

- **Mailing List:** *hw-cwe-special-interest-group-sig-list@mitre.org*
  - ***NOTE: All mailing list items are archived publicly at:***
    - *https://www.mail-archive.com/hw-cwe-special-interest-group-sig-list@mitre.org/*
- **What would members of this body like to see for the next HW SIG agenda?**
- **Questions, Requests to present? Please let us know.**

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Backup

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What is “Completeness?”

- **Conceptual Coverage: CWE corpus has entries for all weakness-related concepts**
  - Minimize gaps (at least to the Base level?)
  - CWEs for weaknesses within scope of the CWE program (e.g., hardware, AI)
- **Individual CWE entries**
  - Entry has all “high-priority” top-level fields (elements) filled
  - Entry has all “preferred” elements filled, where possible
  - Entry has other elements filled as-needed
- **Breadth: all entries have the element**
- **Depth: all elements have full details**
- **Correctness / Quality**

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Completeness Status – as of CWE 4.19 (disclaimer: not all elements are listed)

	All Active Weaknesses			View-1194			MIHW		
Field	**# Filled**	**# Unfilled**	**% Filled**	**# Filled**	**# Unfilled**	**% Filled**	**# Filled**	**# Unfilled**	**% Filled**
Name	944	0	100%	110	0	100%	16	0	100%
Description	944	0	100%	110	0	100%	16	0	100%
Extended_Description	585	359	62%	104	6	94%	15	1	93%
Diagram	75	869	8%	2	108	1%	0	16	0%
Relationships	944	0	100%	110	0	100%	16	0	100%
Common_Consequences	944	0	100%	110	0	100%	16	0	100%
Applicable_Platforms	944	0	100%	110	0	100%	16	0	100%
Modes_of_Introduction	944	0	100%	110	0	100%	16	0	100%
Detection_Methods	493	451	52%	44	66	40%	15	1	93%
Potential_Mitigations	670	274	71%	106	4	96%	16	0	100%
Demonstrative_Examples	592	352	63%	96	14	87%	16	0	100%
Observed_Examples	549	395	58%	51	59	46%	14	2	87%
Weakness_Ordinalities	937	7	99%	110	0	100%	16	0	100%
Likelihood_of_Exploit	183	761	19%	2	108	2%	0	16	0%
References	581	363	62%	78	32	70%	16	0	100%
Mapping_Notes	944	0	100%	110	0	100%	16	0	100%
Terminology_Notes	35	909	4%	2	108	1%	0	16	0%
Alternate_Terms	102	842	11%	2	108	1%	0	16	0%
Related_Attack_Patterns	330	614	35%	91	19	82%	13	3	81%
Relationship_Notes	132	812	14%	6	104	5%	3	13	18%
Taxonomy_Mappings	633	311	67%	11	99	10%	1	15	6%

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Prioritizing Completeness Improvements

- **Q1: Should we choose a specific element, or a set of elements to prioritize for completeness?**
- **Q2: Which CWE(s) do we focus on?**
  - MIHW, all hardware view, category, whatever people are interested in…
- **Q3: Should we focus more on depth than breadth?**
  - Affected languages (and which ones?); more variety in demonstrative examples; effectiveness estimates for mitigations/detection; etc.
- **Q4: How can the HW-SIG help?**
  - What can MITRE do to help facilitate?
    - Detailed reports can be provided. What reporting would be helpful?

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HW CWE SIG: Help Set Our Focus

For the next 3 minutes, please answer these in chat.

Start your answers with Q1 or Q2 so we can sort them later.

PART 1:

- **Q1 – Outcomes (6–12 months)**
  - What is one concrete outcome you'd love this SIG to achieve?
- **Q2 – Themes**
  - What 1–2 HW CWE themes or problem areas are most important for us to tackle?

PART 2:

- **Q3 – Ownership**
  - Is there a theme/outcome you'd be willing to Champion or co-champion?
  - Contribute to in a small working group?

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Current Inputs

Q1 – Outcomes (6–12 months)

- **Improve Data Quality for Next MIHW: How can we encourage more vendors to use HW CWE consistently and correctly?**
- **Identify security risks that are specific to the implementation of functionality by HW (arising from e.g., physics, cost structure, tools and methods for implementation)**

Q2 – Themes

- **"AI for Security": Improve HW CWE collaterals for all LLMs and AI Agents by adding more relevant demonstrative code examples and observed examples to help AI to do their jobs.**
- **"Security for AI": Enrich existing HW CWE contents with more AI related security weaknesses.**

Q3 – Ownership

- **(J. Fung) Demonstrative Code Examples: We plan to continue to partner with Hack@DAC to share more demonstrative code examples for HW CWE this year.**

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